

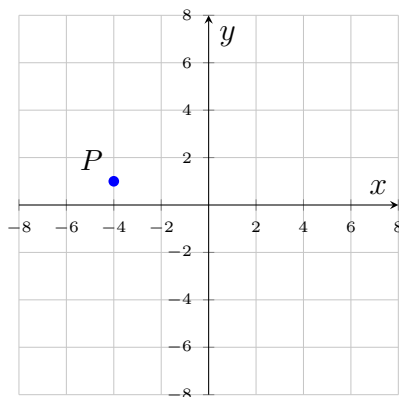
Rigid Motions on the Coordinate Plane

Translating, reflecting and rotating points and segments on the coordinate plane, and checking that lengths do not change

How to use this sheet.

- Every problem gives you a rule in coordinate notation and a grid. Do both: apply the rule in symbols, then plot the image and check that the picture agrees with the algebra. When they disagree, the algebra is usually right and the plotting finger is usually wrong — but look at both before deciding.
- The rules used here, all of them about the origin or the axes: translation $(x, y) \rightarrow (x + h, y + k)$; reflection over the x -axis $(x, y) \rightarrow (x, -y)$; over the y -axis $(x, y) \rightarrow (-x, y)$; over the line $y = x$ $(x, y) \rightarrow (y, x)$; rotation 90° counterclockwise about the origin $(x, y) \rightarrow (-y, x)$; rotation 180° about the origin $(x, y) \rightarrow (-x, -y)$.
- These are all *rigid* motions, which means every distance survives them. On the segment problems you are asked to prove that for one side — compute the length before and after with the distance formula, not with a ruler.

- 1. Translate a point.** $P(-4, 1)$ under the rule $(x, y) \rightarrow (x + 6, y - 3)$.



Apply the rule one coordinate at a time.

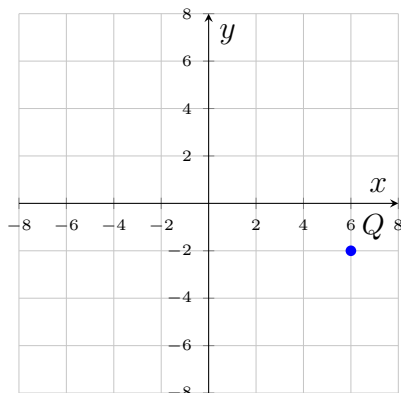
x -coordinate of P' : _____

y -coordinate of P' : _____

So $P' = (\text{_____}, \text{_____})$.

Now plot P' on the grid and label it.

2. Reflect a point over the y -axis. $Q(6, -2)$.



The y -axis is the mirror, so the mirror runs vertically. Which coordinate can a vertical mirror change?

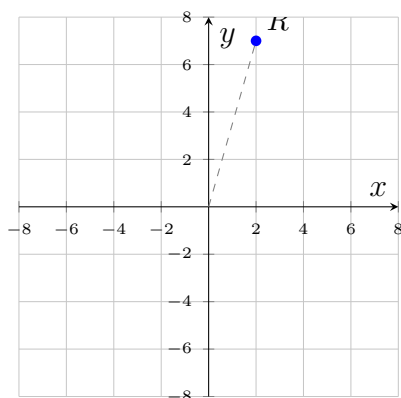
x -coordinate of Q' : _____

y -coordinate of Q' : _____

So $Q' = (\text{_____}, \text{_____})$.

Plot Q' , then check that Q and Q' are the same distance from the y -axis.

3. Rotate a point 90° counterclockwise about the origin. $R(2, 7)$.



The rule is $(x, y) \rightarrow (-y, x)$: the coordinates swap and then the new first one changes sign.

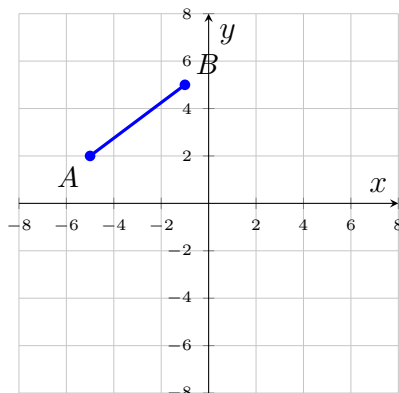
x -coordinate of R' : _____

y -coordinate of R' : _____

So $R' = (\text{_____}, \text{_____})$.

Plot R' and draw the dashed spoke from the origin to it. The two spokes should look perpendicular — and be the same length.

4. Translate a whole segment. $A(-5, 2)$ and $B(-1, 5)$ under $(x, y) \rightarrow (x + 4, y - 6)$.



$$A' = (\quad, \quad)$$

$$B' = (\quad, \quad)$$

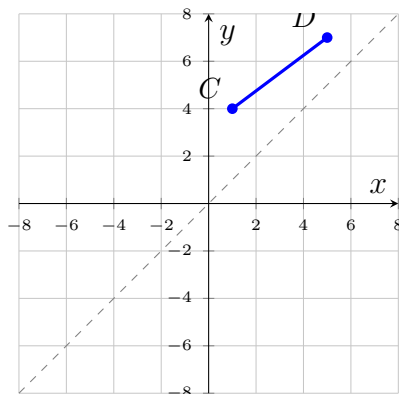
Draw $\overline{A'B'}$ on the grid.

Now use the distance formula on your image points:

$$A'B' = \underline{\hspace{2cm}}$$

Compare it with AB , which you can also compute. A translation slides everything by the same amount, so the two lengths must agree.

5. Reflect a segment over the line $y = x$. $C(1, 4)$ and $D(5, 7)$.



The dashed mirror is the line $y = x$. Its rule swaps the coordinates.

$$C' = (\quad, \quad)$$

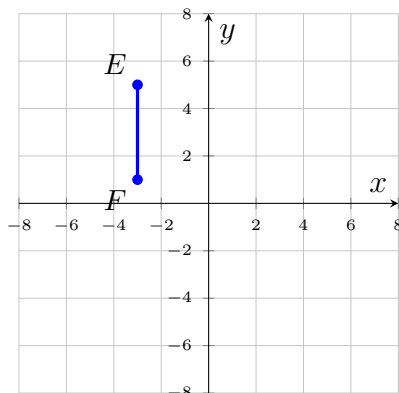
$$D' = (\quad, \quad)$$

Draw $\overline{C'D'}$.

$$C'D' = \underline{\hspace{2cm}}$$

A check worth doing: the mirror should cut $\overline{CC'}$ in half and at a right angle.

- 6. Rotate a segment 180° about the origin.** $E(-3, 5)$ and $F(-3, 1)$.



A half turn about the origin negates both coordinates.

$$E' = (\quad, \quad)$$

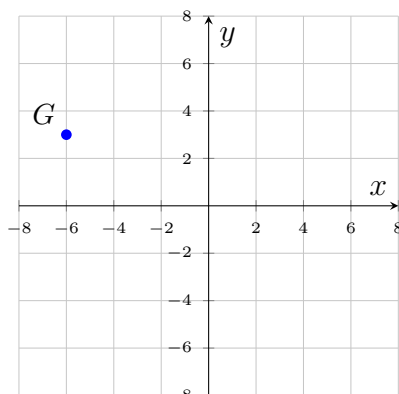
$$F' = (\quad, \quad)$$

Draw $\overline{E'F'}$.

$$E'F' = \quad$$

Which quadrant did the segment start in, and which did it land in? A half turn always moves a figure to the diagonally opposite quadrant.

- 7. Find and fix the mistake.** Sam is asked to reflect $G(-6, 3)$ over the y -axis. Sam writes $G'(-6, -3)$.



Which transformation did Sam actually perform?

The correct image:

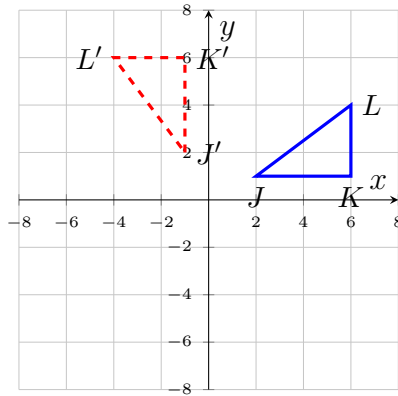
x -coordinate: _____ y -

coordinate: _____

So $G' = (\quad, \quad)$.

Plot G , Sam's answer, and the correct image. One of the two lies on the wrong side of the wrong mirror.

8. Name the motion, then justify the congruence. Triangle JKL has been mapped onto triangle $J'K'L'$ by a single rigid motion.



(a) Name the motion in words, and write its coordinate rule.

(b) Show your rule sends each of J , K and L to the right image point.

(c) Explain how you know $\triangle JKL \cong \triangle J'K'L'$ *without* measuring all three pairs of sides.